

NAKSHATRA-X : Full AI Roadmap

A Simple Guide to Our Tools & How They Work

Our Goal

Make mining safer and more profitable using real-time AI and satellite data, without making things complicated.

The Tools We Use & How They Work

1. Open-Meteo & ISRO Bhuvan (The Eyes)

What it is: Live satellite and weather data streams.

How we use it: It constantly watches the exact GPS location of MOIL mines. If heavy rain is coming, it tells the AI immediately so we can stop floods before they ruin the haul roads.

2. XGBoost & Prophet (The Brain)

What it is: Powerful Machine Learning (AI) models.

How we use it: These models have memorized 50 years of mining history. They look at today's real weather from Open-Meteo and predict exactly how much manganese we will fail to mine next week if we don't take action today.

3. SciPy Simplex Optimizer (The Calculator)

What it is: Advanced mathematical solver.

How we use it: Customers want 41% pure Manganese. We have different piles of ore (some 30%, some 50%). SciPy instantly calculates the cheapest and fastest way to mix these piles to hit the exact 41% target.

4. IndexedDB & WebCrypto (Offline Security)

What it is: Secure browser storage.

How we use it: Mines often lose internet. We save an encrypted copy of the AI brain directly onto the manager's laptop browser. If the internet dies, the dashboard keeps running and predicting locally!

5. Next.js & React (The Face)

What it is: Modern website builders.

How we use it: This is what the manager sees. It turns all the complex math and satellite data into simple, beautiful buttons and graphs that anyone can understand.

Future Roadmap

- **Phase 1 (Now):** Predicting weather risks and mixing ores automatically.
- **Phase 2 (Next Year):** Connecting real IoT sensors from the mining trucks directly to the AI to predict when a truck engine will break.
- **Phase 3 (Future):** Fully automated drone dispatches for 3D mapping of the mine every morning.